

Learning what a quantum sensor can reveal



From a trainable quantum circuit
to the limits of an experimental record.

Luca Manzi Artificial Intelligence for Science and Technology

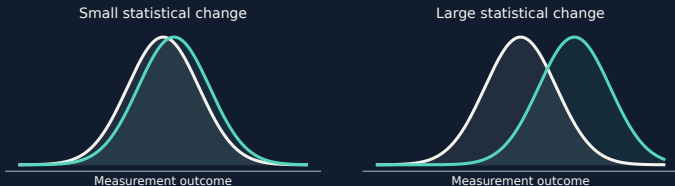
Supervisor: Prof. Enrico Prati Co-supervisor: Dr. Sebastiano Corli

Thesis: Estimation of Quantum Fisher Information by variational quantum circuits
on a quantum computer

THE QUESTION

Can we distinguish two nearby worlds?

A slightly different magnetic field or planet position should change what we observe.



Fisher information

How sensitive the observed statistics are to a small parameter change.

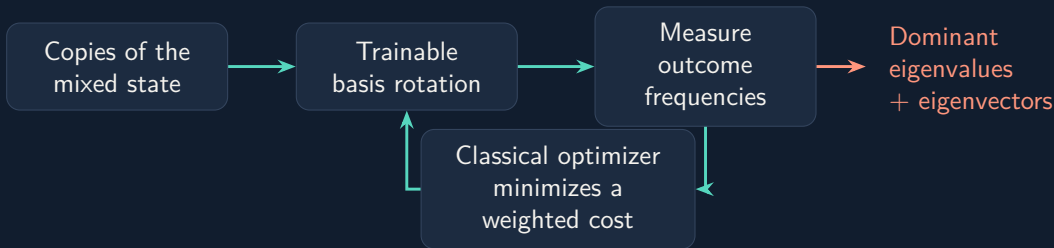
Quantum Fisher information (QFI) is the best local information available from the state, optimized over measurements.

Illustrative distributions. QFI is a local benchmark; attainable precision also depends on the measurement and estimation assumptions.

Learn the dominant components

A variational route to mixed-state information bounds.

An n -qubit density matrix has $2^n \times 2^n$ entries.
Dense classical diagonalization scales as $\mathcal{O}(8^n)$.

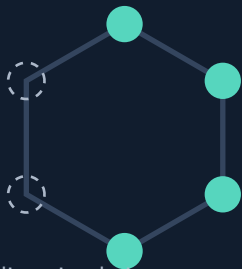


A learned basis supplies the spectral information needed for QFI bounds.

VQSE / VQFIE implementation. Established algorithms: Cerezo et al. (2022); Beckey et al. (2022).

A sensor is more than its purest state

Six interacting spins encode a magnetic field. Only four are accessible.



Periodic spin chain

4 retained / 2 discarded

Ground state Lowest-energy preparation

Depolarized Fixed depolarizing noise

Thermal Field-dependent populations

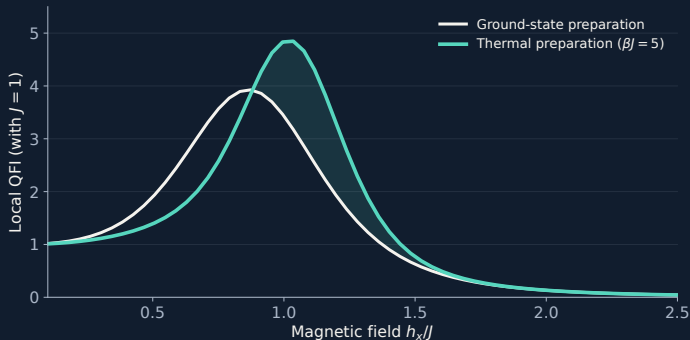
Compare preparation families over the operating field range.

Transverse-field Ising model (TFIM), $N = 6$, $n = 4$, periodic boundaries, $J = 1$. Thesis: NoiseAnalysis, Project 9.

RESULT 1 / PREPARATION MATTERS

Thermal preparation can be more informative

A finite-system advantage over the reduced ground-state family.



+23.5%

higher peak local QFI

40/60

sampled fields favour
thermal preparation

Optimize for the operating regime, as well as the peak.

Periodic $N = 6$, $n = 4$, $\beta J = 5$; 60 fields on $h_x/J \in [0.1, 2.5]$. Reproduced from the thesis model; exact SLD calculation.

The measurement decides what becomes visible

A faint planet produces a tiny change beside a bright star.

Direct imaging



Record where each photon lands.

Spatial-mode sorting



Record which spatial mode is occupied.

SPADE changes the measurement basis to reveal weak spatial signals.

Schematic, not experimental imagery. SPADE: spatial-mode demultiplexing. Tsang et al. (2016); Santamaria et al. (2024).

THE EXPERIMENTAL TURN

Many observations. An incomplete view.

The completed ASI Matera experiment supplied aggregate first-order counts.

52,500

count observations across 525 settings
100 repetitions per setting, 10 ms each

Predict the observed response.
Quantify its variability.

Missing observables

Separate modal outputs

Measurements in other bases

Incident flux and failure outcomes

The record supports prediction, but cannot reconstruct the quantum state.

Thesis: ExoplanetExperiment, dataset description. No programmable basis rotations were recorded; experimental TQFI was not obtained.

Connect the physics to the recorded data

A sequential model makes the assumptions and information losses explicit.



Quantum states
populations *and* coherences

Classical probabilities
calibration and count statistics

Physical constraints + learned calibration + explicit uncertainty

A good fit constrains the observed response, not every hidden mechanism.

Thesis: sequential quantum-to-classical framework. Optical loss and inaccessible modes retain explicit failure outcomes in the model.

RESULT 2 / HELD-OUT PREDICTION

Physics + calibration predicts unseen settings

A Gaussian-process correction captures smooth setting-dependent response.

$$\underbrace{\mu(d_a, \epsilon)}_{\text{setting mean}} = \underbrace{\mu_0(d_a, \epsilon)}_{\text{physical response}} \underbrace{\exp[f(d_a, \epsilon) + \xi]}_{\text{calibration correction}}$$

Checkerboard cells

93.5%

95% interval coverage

Distance rows

94.7%

95% interval coverage

Source-ratio columns

93.7%

95% interval coverage

Grouped five-fold validation; all model parameters refitted within each training fold.

Validated target: the 100-repeat setting mean.

Thesis: ExoplanetExperimentAppendix, hierarchical calibration table. Rounded coverage; standardized residual SD 1.00–1.04.

Readout can erase the theoretical advantage

Compare information checkpoints using the same source model.

Resolved low-mode outcomes

99.68–100%

of source-state QFI retained
as classical Fisher information

Binary readout + cross-talk

0.13–57.0%

under an assumed symmetric
0.35% port-confusion model

Measurement design and readout belong inside the sensing problem.

Same nine nonzero settings; per photon surviving common loss. Ranges across settings, not confidence intervals. Conditional model.

TAKE-HOME MESSAGE

Useful sensing is an end-to-end problem

Learn the representation. Match the preparation. Model the observation.

01 Learn a compact state representation
Implemented variational spectral estimation for QFI bounds.

02 Choose the preparation for the task
Thermal populations can improve local sensitivity in the tested regime.

03 Validate what the data can support
Predictive calibration and explicit limits on latent-state inference.

Next: an experiment with resolved outputs and a trainable optical basis.

What exactly is bounded?

The finite-displacement diagnostic and the local QFI are distinct.

$$\mathcal{I}_{\Delta} = \frac{8[1 - F(\rho_{\theta}, \rho_{\theta+\Delta})]}{\Delta^2}$$

Root fidelity F compares nearby states.

Dominant-eigenspace truncation bounds this fidelity quantity.

At regular points, $\mathcal{I}_{\Delta} \rightarrow \mathcal{F}_Q$ as $\Delta \rightarrow 0$.

Finite-step bounds are not automatically pointwise bounds on exact SLD QFI.

Methods: finite-displacement TQFI / SSQFI bounds. The fidelity limit requires regularity at the operating point.

Why does this respect data processing?

Different preparation families can have different field responses.

$$\rho_{\beta}(h) = \frac{e^{-\beta H(h)}}{\text{Tr } e^{-\beta H(h)}}$$

The field changes eigenvectors *and* thermal populations.

$$\mathcal{F}_Q[\text{Tr}_E \rho_{\beta}(h)] \leq \mathcal{F}_Q[\rho_{\beta}(h)]$$

A thermal preparation is not a fixed channel applied to the ground state.

The advantage compares preparations; it does not reverse a fixed noise channel.

Thesis: NoiseAnalysis, Project 9. Exact finite-system SLD benchmark, not a universal temperature advantage.

Generalization across the setting grid

Target: the mean of 100 repeated counts, in counts per 10 ms.

Split	RMSE	Standardized SD	95% coverage
Checkerboard	0.606	1.004	93.52%
Distance rows	0.734	1.041	94.67%
Source-ratio columns	0.609	1.009	93.71%

Matérn-3/2 GP: smooth calibration + independent setting scatter.
 Individual counts: median Fano factor $\simeq 2.50$ (overdispersed).

Coverage is evidence about predictive uncertainty for means, not channel identification.

Thesis: ExoplanetExperimentAppendix, hierarchical calibration validation and raw-count diagnostics.

What would make VQ-SPADE possible?

A proposed optical realization of variational state eigensolving.

- 1 A programmable coherent transformation of spatial modes.
- 2 Resolved modal counts, normalization and failure outcomes.
- 3 Additional overlap measurements for nearby states.
- 4 Independent flux, background and channel calibration.

A future acquisition would turn the learning loop into an optical protocol.

Thesis: proposed VQ-SPADE scheme. Not implemented or tested in the completed experiment.